

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

Denis Akhiyarov

Sr Staff Research Scientist at ServiceNow

[@den-run-ai](#) · github.com/den-run-ai/askme

AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one native tool call → execution evidence ↻

01

CONTROLLABLE SMALL LLMS

Choose the model boundary

- Execution speed
- Hardware and deployment
- Post-training access

02

ASKME LOOP

One small action at a time

```
plan → act → execute / test  
→ fresh evidence ↻
```

Interpret the result. Update locally when possible.

03

EXTERNAL ACCEPTANCE LAYER

Check the delivered workflow

- Preserve the full contract
- Check required behavior
- Accept the real artifact independently

The bridge: eight native tools + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c  
build ./main  
run ./main  
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

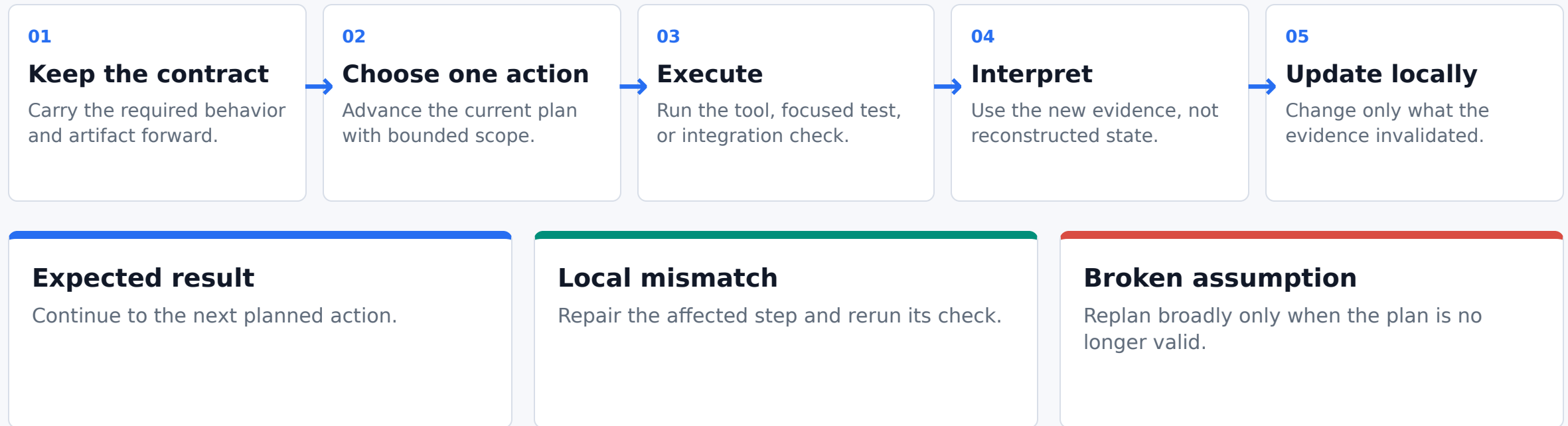
From the agent's side, the task looked done.

Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion $\neq$ accepted artifact.

Hypothesis: update only what fresh evidence invalidates



Design goal: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

Acceptance caught the one bad deliverable

Four hosted models · two simple tasks · one run each

8 / 8

reported complete

7 / 8

artifacts accepted

The fastest build was the rejected one.
Completion and speed were not enough.

HOSTED VARIANT	ARTIFACT BUILD	SANITY REPAIR
Gemma 4 26B A4B	Accepted	Accepted
Gemma 4 31B	Accepted	Accepted
Qwen3.6-27B	Accepted	Accepted
Qwen3.6-35B-A3B	Wrong path	Accepted

Compatibility smoke, not a ranking: single runs on simple tasks — no model comparisons.

Applied patches, unresolved feature task

After bundled harness changes and a changed serving stack.

BEFORE · JULY

0 writes

Empty patch

Revision 2 left both cells empty. Their failure modes differed.

AFTER · AUG 1

Patches applied

→ **11 / 13** **7 / 13**

Gemma target tests Qwen target tests
Held-out target tests passed in part;
both patches remained unresolved.

GAPS IN BOTH ATTEMPTS

No target-test run

→ Gemma rewrote one file 18 times.
Qwen returned to reading. Both agents
exhausted their planning attempts.

Observed: partial target-test acceptance, without agent completion.

One task, one attempt per model · historical results · no causal attribution or benchmark score.

Promising for bounded loops. Feature readiness is still open.

Observed

Historical smoke: 7 / 8 artifacts accepted.
Feature canary: applied but unresolved patches; 11/13 and 7/13 target tests.
Neither ran target tests; both agents exhausted.

Supported

Action success, agent completion, and accepted artifacts are distinct. The changed outcomes do not isolate a harness effect.

Still open

Reliable feature work, local performance, and the benefit of each design choice. One task and changed configurations cannot settle them.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries · documentation checked Sep 8, 2026.

	AskMe	pi	OpenHands
ACTION SURFACE	8 native tools: 6 executable actions + <code>done</code> / <code>fail</code> ; one call per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible <code>Action</code> → <code>Observation</code> tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	<code>done</code> + conditional validation. Unavailable check: <code>complete_unverified</code> . Held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	<code>finish</code> signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.